17

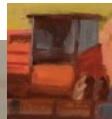
# Production and Growth

PowerPoint Slides prepared by:
Andreea CHIRITESCU
Eastern Illinois University

© 2011 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

# Surprise Test!

- You have 15 mins to answer 5 questions.
- **I will not accept any answer sheet after the time specified.**
- Please write only short, to the point answers.
- Weightage: 2 marks; 0.4 marks for each question.
- Please try not to cheat. It will be hard for me to give you ZERO on your very first exam.



# Questions

Define Gross Domestic Product.

Mention the components of GDP (spending) and briefly explain each.

What is the difference between Nominal GDP and Real GDP.

Explain the two differences between CPI and GDP Deflator as measures of inflation.

Stock or Flow?: GDP; Unemployment; Government Debt. Explain briefly.

# Economic Growth

- Real GDP per person
  - Living standard
  - Vary widely from country to country
- Growth rate
  - How rapidly real GDP per person grew in the typical year
- Because of differences in growth rates
  - Ranking of countries by income changes substantially over time (Jpn, Brz, Can vs UK)
  - Compounding effect of rate of growth
    - 6%-18x, 8%-47x, 10%-117x (in 50 years)

© 2011 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

# Table 1

## The Variety of Growth Experiences

| Country | Period | Real GDP per Person at Beginning of Period[a] | Real GDP per Person at End of Period[a] | Growth Rate (per year) |
|---|---|---|---|---|
| Japan | 1890–2008 | $1,504 | $35,220 | 2.71% |
| Brazil | 1900–2008 | 779 | 10,070 | 2.40 |
| Mexico | 1900–2008 | 1,159 | 14,270 | 2.35 |
| Germany | 1870–2008 | 2,184 | 35,940 | 2.05 |
| Canada | 1870–2008 | 2,375 | 36,220 | 1.99 |
| China | 1900–2008 | 716 | 6,020 | 1.99 |
| United States | 1870–2008 | 4,007 | 46,970 | 1.80 |
| Argentina | 1900–2008 | 2,293 | 14,020 | 1.69 |
| United Kingdom | 1870–2008 | 4,808 | 36,130 | 1.47 |
| India | 1900–2008 | 675 | 2,960 | 1.38 |
| Indonesia | 1900–2008 | 891 | 3,830 | 1.36 |
| Pakistan | 1900–2008 | 737 | 2,700 | 1.21 |
| Bangladesh | 1900–2008 | 623 | 1,440 | 0.78 |

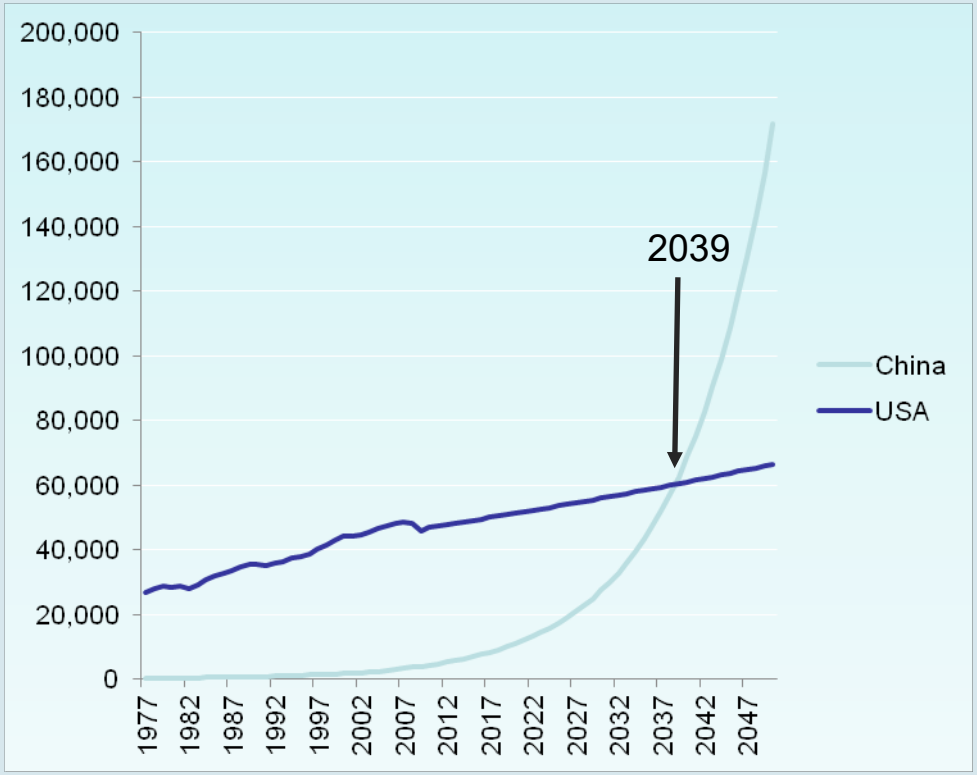
[a]Real GDP is measured in 2008 dollars.

© 2011 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

# A Tale of Two Nations – Inside Edition

- In 2010, GDP per capita was US$ 47,032 for the U.S. and US$ 4,364 for China.
- An average person in the U.S. currently earns about 10 times that an average person earns in China.
- Between 2001 and 2010, the U.S. grew at about 1% per year on average while China grew at 10% per year on average.
- If incomes in both countries continue to grow at the pace of the past decade, how long will it take China to surpass the U.S. in GDP per capita?
- *How long will China continue to grow at 10%?*

# What if the current growth paths continue…




Hot Debate: How likely will this happen? What will make it happen?
What drives economic growth?

# Productivity

- Productivity
  - Quantity of goods and services
  - Produced from each unit of labor (work hour)
  - More work done per unit of time → more production (even with less amount of labor force)
- Why productivity is so important
  - Key determinant of living standards
  - Growth in productivity is the key determinant of growth in living standards

© 2011 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

# Productivity

- Determinants of productivity
  - Physical capital
    - Stock of equipment and structures
  - Human capital
    - Knowledge and skills that workers acquire through education, training, and experience
  - Natural resources
    - Nature, such as land, rivers, and mineral deposits
  - Technological knowledge
    - Knowing the best ways to produce

© 2011 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

# Saving and Investment

- How to raise future productivity
  - Invest more current resources in the production of capital
    - Optimal Saving?
  - Trade-off
    - Devote fewer resources to produce goods and services for current consumption
    - *Higher savings rate more important indicator of long run growth*.

© 2011 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

# Diminishing Returns

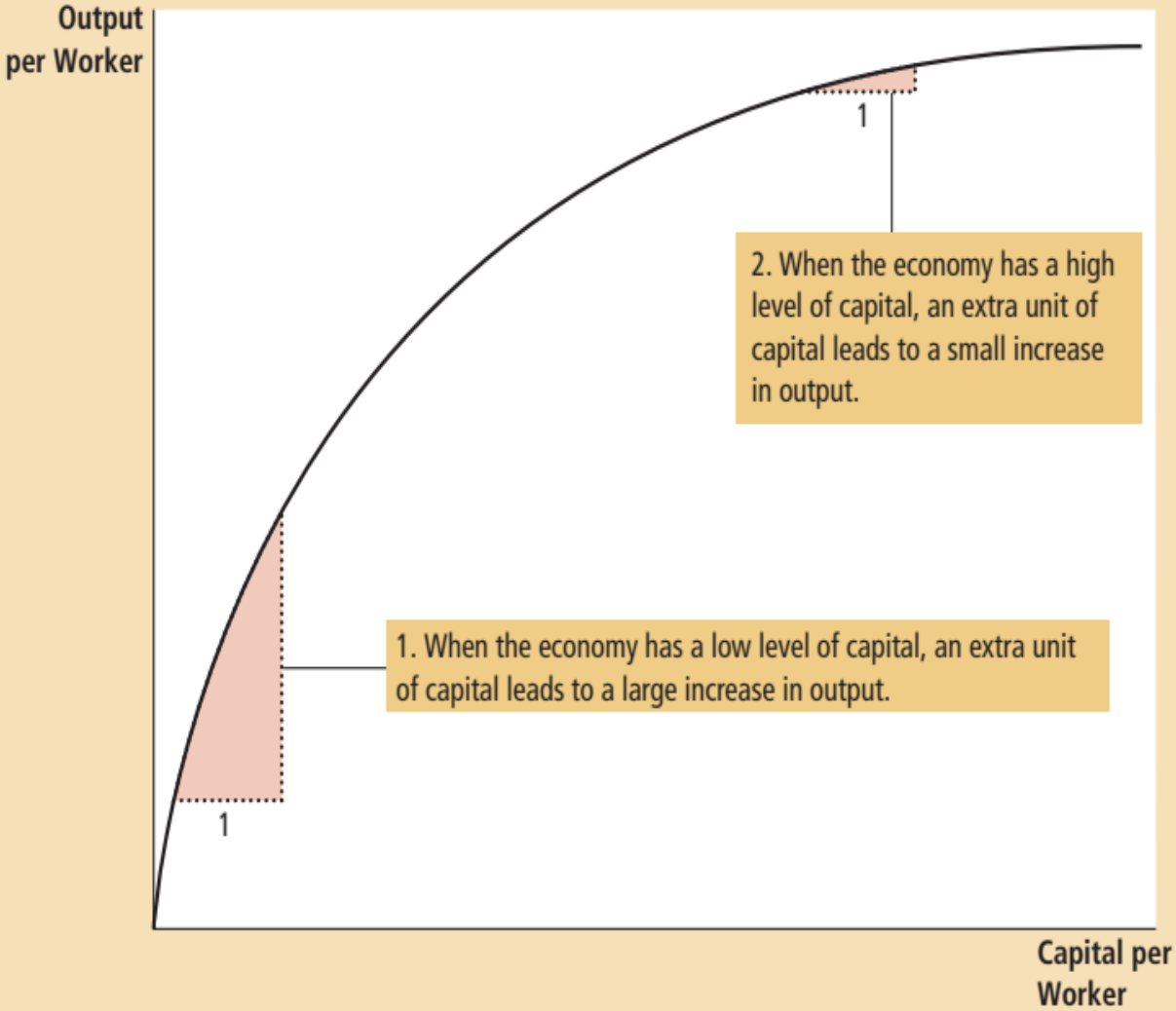
- Higher savings rate
  - Fewer resources used to make consumption goods
  - More resources to make capital goods
  - More capital stock increases productivity
  - More rapid growth in GDP
  - Diminishing returns set in as more and more capital is accumulated
  - Growth slows down eventually

© 2011 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Figure 1

**Illustrating the Production Function**

This figure shows how the amount of capital per worker influences the amount of output per worker. Other determinants of output, including human capital, natural resources, and technology, are held constant. The curve becomes flatter as the amount of capital increases because of diminishing returns to capital.



# Catching Up

- Catch-up effect
  - Countries that start off poor tend to grow more rapidly than countries that start off rich
  - Japan, Germany after WWII
- Poor countries
  - Low productivity
  - Even small amounts of capital investment
    - Can increase worker productivity substantially
    - So they grow faster with the same increase in investment

© 2011 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

# Investment from Abroad

- Benefits from foreign investment
  - Increase the economy's stock of capital
  - Higher productivity
  - Higher wages
  - State-of-the-art technologies from advanced countries

© 2011 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

# Education, Health and Nutrition

- Education creates skilled labor force
  - positive externalities
- Vicious circle in poor countries
  - Poor countries are poor
    - Because their populations are not healthy
  - Populations are not healthy
    - Because they are poor and cannot afford better healthcare and nutrition
  - Many countries in Africa
  - again positive externalities

© 2011 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

# Property Rights, Political Stability

- Government to foster economic growth
  - Protect property rights
    - Private ownership of resources
    - Needed for markets to work
    - Corruption hampers enforcement of property rights
  - Promote political stability
    - Riots & wars threaten property rights

© 2011 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

# Free trade and Population Growth

- Free trade:
  - comparative advantage → specialization
  - Inward vs outward outlook
- Does more population help growth?
  - Larger population means more workers (more skilled labor?), but also more consumers
  - Thinning out available resources & capital
    - Lower productivity and income
  - Poorer countries tend to have higher population growth

© 2011 Cengage Learning. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.